

$$Z'_i = \frac{0.968 \times 10^3 \times 9.1 \times 10^3}{9.1 \times 10^3 + 0.968 \times 10^3} = \frac{8808.8}{10.068} = 875 \, \Omega$$

$$= R_i \rightarrow \left[\because Z'_L = \frac{Z_{in} \cdot R_B}{(Z_{in} \cdot R_B)} \right]$$

$$A_{vs} = \frac{V_o}{V_s} = \frac{V_o}{V_i} \times \frac{V_i}{V_s} = \frac{V_o}{V_i} \times \frac{R_i}{R_s + R_i}$$

$$= A_v \times \frac{R_i}{R_s + R_i} = \frac{-83 \times 0.875 \times 10^3}{1 \times 10^3 + 0.875 \times 10^3} \approx -39$$

A_{vs} refers to voltage gain taking the signal source resistance R_s into consideration and A_{is} refers to current gain taking R_s into consideration.

$$A_{is} = A_i \times \frac{R_s}{R_s + R_i} = -48 \times \frac{10^3}{1.875 \, K} = 25.6 \approx -26$$

SOLVE. For a single-stage Transistor Amplifier $R_s = 1 \, k\Omega$ and $R_L = 10 \, k\Omega$. The h-parameter values are $h_{ie} = 50$, $h_{ie} = 1.1 \, k\Omega$, $h_{re} = 2.5 \times 10^{-4}$ and $h_{oe} = 25 \, \text{mhos}$. Find the values of A_i , A_v , A_{vs} , A_{is} , power gain A_p , R_i , R_o for the CE transistor configuration.

UNIT TWO

SEMICONDUCTOR DEVICES RELATED TO ELECTRICAL POWER SYSTEM

1.0 Introduction

Thyristors, Diacs and Triacs are semiconductor solid-state devices specially developed for controlling power in AC circuits or in electrical power systems. They provide better control of circuit current than other electrical devices like rheostats which controls current through its resistance. Thyristors, Diacs and Triacs use the principle of switching (or voltage control) to control the flow of current to electrical loads and are very efficient because, they are very fast in action and dissipate very little power when they are open or closed. This unit describes the construction, principles of operation and uses of Thyristors, Diacs and Triacs.

1.1 THYRISTOR (Silicon Controlled Rectifier)

Thyristor as Four-Layer Semiconductor

When a P-N junction is added to an N-P-N transistor structure, the device formed is commonly called a silicon controlled rectifier (SCR) or thyristor. Fig. 15(a) shows the construction. It is a silicon P-N-P-N or four-layer diode with three P-N junctions; J_1 , J_2 and J_3 having three terminals connected to the semiconductor materials called anode (A), gate (G) and cathode (K) as shown in Fig. 15a with its circuit symbol shown in Fig. 15b. The gate terminal is connected to the cathode of the diode symbol. The diode arrow points in the direction of forward current or majority carriers through the thyristor. It is called a diode because it conducts in one direction and blocks in the other direction.

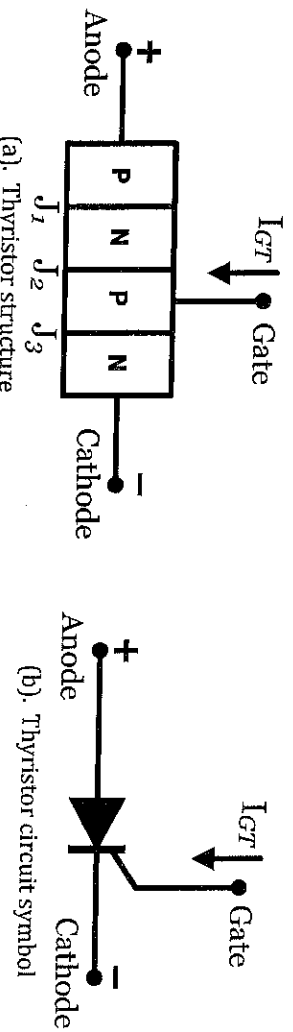


Fig.15. Three-terminal, four-layer diode or SCR

The voltage applied to the Gate controls the conduction of the thyristor. For proper operation of the thyristor, the anode terminal is connected to a highly positive voltage

than the Cathode and the Gate is connected to a small positive voltage than the Cathode. A thyristor is therefore used to control large amounts of power by using small amount of power in the gate circuit. It is widely used as a switching device in power control applications. It can control loads by switching the current OFF and ON up to many thousand times a second and can also be switched ON for variable lengths of time, thereby delivering selected amount of power to the load. Hence, a thyristor combines the characteristics of a rectifier and a transistor or rheostat and a switch.

Why SCR called Thyristor? The SCR is commonly called a thyristor because it is a semiconductor equivalent of a thyatron. The Gate, Anode and Cathode of SCR correspond to the Grid, Plate and Cathode of the thyatron.

Basic Thyristor (SCR) Operation

In a thyristor, the load is always connected in series with the anode. The anode is always connected to the positive supply and the cathode to the negative supply. The operation of a thyristor can be described under two conditions; when the gate is open and when it is connected to positive supply.

(1) When Gate is open

Under this condition, no voltage is applied to the Gate. Current flow to the load, R_L is blocked because the junction J_2 is reverse-biased while J_1 and J_3 are forward-biased. The thyristor is described as **cut off**. But, when the anode voltage is increased, a particular critical value called **Forward Breakover Voltage (V_{BO})** is reached when J_2 breaks down and the thyristor suddenly switches ON and begins to conduct. It is said to be in the **ON state**. The forward resistance of the thyristor becomes very low ($0.01\ \Omega - 1\ \Omega$) and current keeps flowing continuously until the circuit is opened. This is shown in Fig. 16.

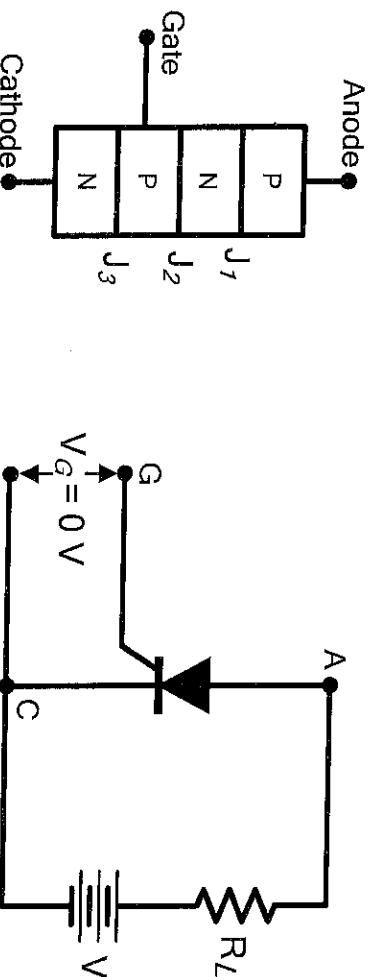


Fig. 16. Thyristor operation when Gate is open

(2) When Gate is connected to small positive voltage

Under this condition, a small positive voltage is connected to the Gate and current flow through the thyristor starts to increase heavily. Junctions J_1 and J_3 are forward-biased while J_2 is reverse-biased. The electrons from the n-type material at the cathode move across the junction J_3 whereas holes from the p-type move toward the cathode. Also, the electrons from J_3 are attracted across the reverse-biased J_2 because of the small positive Gate voltage. Thus, the gate current starts flowing. As soon as the gate current flows, anode current starts to increase. The increased anode current in turn makes more electrons to gather at J_2 . This process will continue and after a very short time, J_2 will break down and the thyristor starts to conduct heavily.

Immediately the thyristor starts conducting, the Gate loses all control. Even if the Gate voltage is removed, the anode current does not decrease at all. The only way to stop conduction (turn off the thyristor) is to reduce the applied voltage to zero.

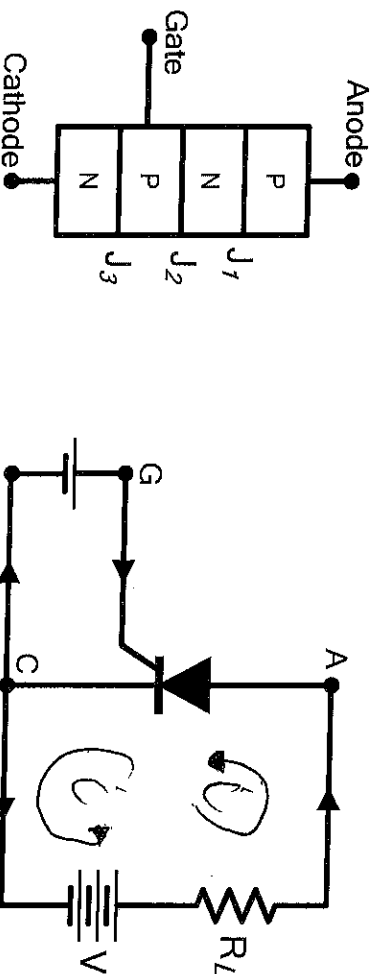


Fig. 17. Operation with Gate connected to positive voltage

Summary of Thyristor Operation

- 1) A thyristor has two states: either ON (conducts heavily) or OFF (does not conduct). Therefore, a thyristor behaves like a switch.
- 2) There are two ways to turn ON a thyristor. The first method is to keep the Gate open and make the supply voltage, V equal to the breakover voltage V_{BO} . The second method is to operate thyristor with supply voltage less than V_{BO} and then turn it on by connecting a small positive voltage (about 1.5 V, 30 mA) to the Gate.
- 3) Connecting a small positive voltage to the Gate is the normal way to turn ON a thyristor because V_{BO} is usually much greater than supply voltage, V .
- 4) To turn OFF a thyristor, reduce the supply voltage to zero.